# Introduction to Artificial Intelligence

Semester I, 2025-26

## Modeling Uncertainty

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# Outline

- Last Class
  - Informed Search
- This Class
  - Modeling Uncertainty
- Reference Material
  - AIMA Ch. 13 and 14

# Acknowledgement

**These slides are intended for teaching purposes only. Some material has been used/adapted from web sources and from slides by Doina Precup, Dorsa Sadigh, Percy Liang, Mausam, Dan Klein, Anca Dragan, Nicholas Roy and others.**

# Uncertainty is Pervasive in AI

- Models till now assumed that everything was certain
- Exact knowledge of the agent's domain
- Actions had precise effects - they acted as they were intended.
  - i.e., if you are in state S1 and you execute action A you arrive at state S2.

- In reality, we must have the ability to deal with uncertainty.
  - We might not know the status of a road when we choose to drive down it.
  - We don't know what the patient may be suffering from.

- We need ways to model uncertainty and incorporate uncertainty in our decision making.
  - We try to leave earlier than usual if it has rained.

# Uncertainty in the real world

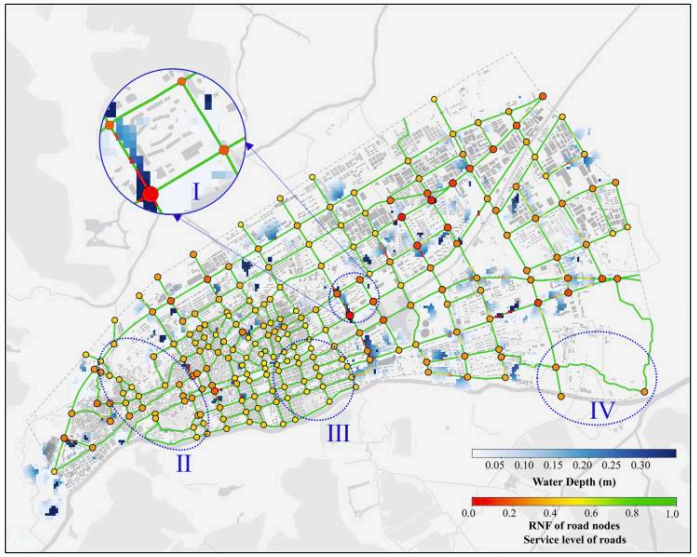
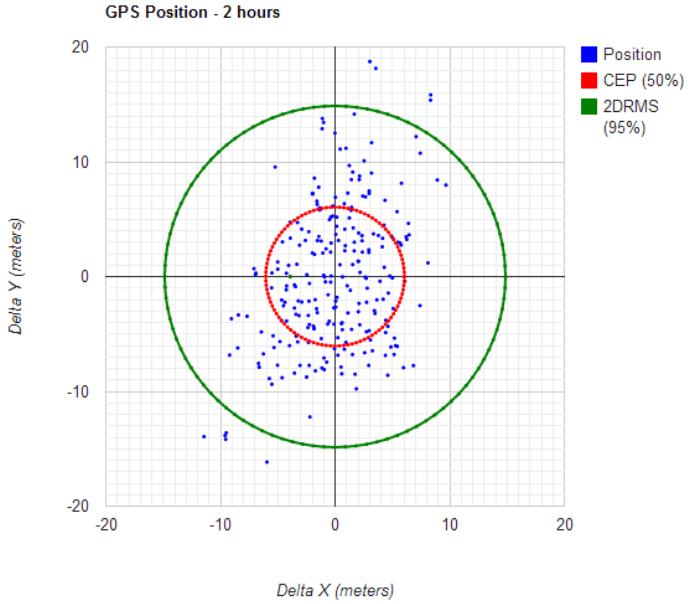
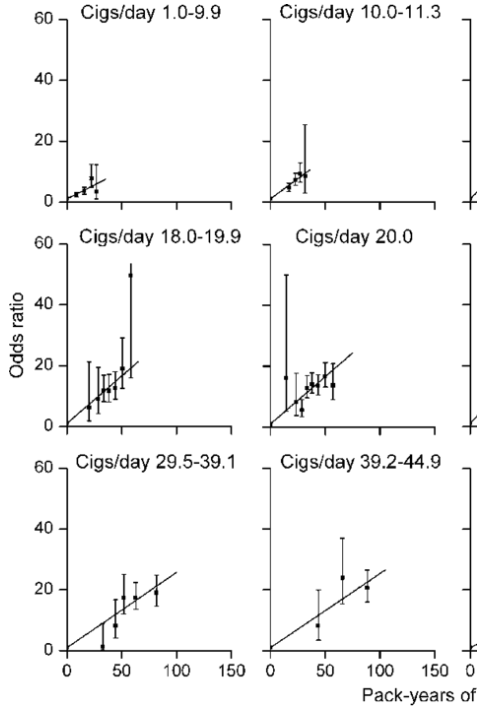
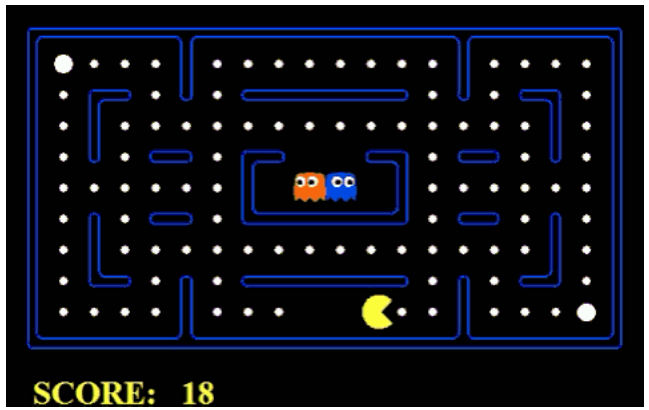


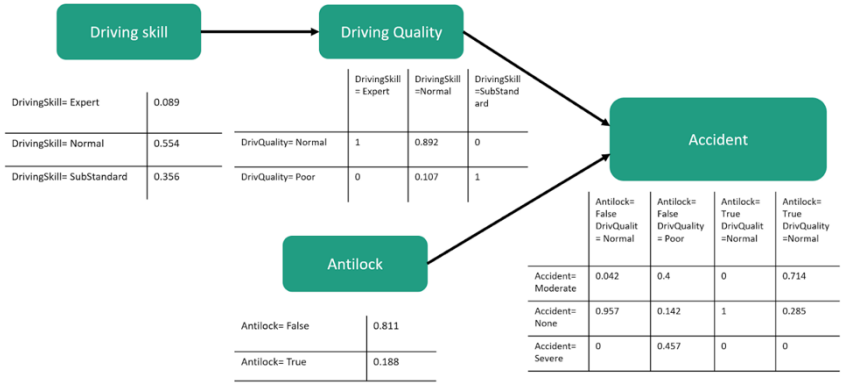

**Fig 6.** An inundation instance and concurrent RNF in Ling-Xi.





# Random Variables

- A random variable is some aspect of the world about which we (may) have uncertainty
  - R = Do I have Covid?
  - T = Engine is faulty or working?
  - D = How long will it take to drive to IIT?
  - L = Where is the person?

- Domains
  - R in {true, false}  (often write as {+r, -r})
  - T in {faulty, working}
  - D in [0, ∞)
  - L in possible locations in a grid {(0,0), (0,1), …}

**I hear an unusual sound and a burning smell in my car, what fault is there in my engine?**

**I have fever, loss of smell, loss of taste, do I have Covid?**

**I hear some footsteps in my house, where is the burglar?**

# Probability Distributions over Finite Sets

- A probability is a function defined over a set of atomic events U.
- U represents the universe of all possible events.
- Given U (a universe of events), a probability function is a function defined over subsets of U that maps each subset onto the real numbers
  - that satisfies the Axioms of Probability.

**1. $P(U) = 1$**

**2. $P(A) \in [0,1]$**

**3. $P(\{\}) = 0$**

**4. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$**

***NB: if $A \cap B = \{\}$ then $P(A \cup B) = P(A) + P(B)$***

# Probability Distributions over Finite Sets

$A \vee B$: to represent the set of events with either property A or B, i.e. the set $A \cup B$

$A \wedge B$: to represent the set of events both property A and B, i.e. the set $A \cap B$

$\neg A$: to represent the set of events that do not have property A: the set U-A (i.e., the complement of A w.r.t. the universe of events U)

# Probability Distributions over Finite Sets

We will model sets of events in our universe as vectors of feature values.

these are the constraint satisfaction problem

Like CSPs, we have

1. a set of variables $V_1, V_2, \ldots, V_n$

2. a finite domain of values for each variable, $Dom[V_1], Dom[V_2], \ldots, Dom[V_n]$.

The universe of events U is the set of all vectors of values for the variables

$$\langle d_1, d_2, \ldots, d_n \rangle : d_i \in Dom[V_i]$$

When we write P(A=a, B=b), we will mean the probability that variable A has been assigned value 'a' **and** variable B has been assigned value 'b'. Note that here, sets of events are induced by a given value assignment. So, P(A=a) represents a set of events in which A holds the value 'a'.

# Probability Distributions over Finite Sets

Our event space has size $\prod_i |Dom[V_i]|$, i.e., the product of the domain sizes. If $|Dom[V_i]| = 2$, we have $2^n$ distinct atomic events.

Note the size of possible event outcomes (or variable assignments) grows **exponentially** with the number of variables.

# Probability Distributions over Finite Sets

We often want to look at subsets of U defined by value assignments to particular variables.

E.g.

**$\{V_1 = a\}$ is the set of all events where $V_1 = a$**

**$\{V_1 = a, V_3 = d\}$ is the set of all events where $V_1 = a$ and $V_3 = d$.**

# Probability Distributions over Finite Sets

If we have probability of every atomic event (wherein every event is a full instantiation of the variables) we can compute the probability of any other set of events.

E.g.

**$\{V_1 = a\}$ is the set of all events where $V_1 = a$**

**$P(\{V_1 = a\}) =$**

$$\sum_{x_2 \in Dom[V_2]} \sum_{x_3 \in Dom[V_3]} \sum_{x_4 \in Dom[V_4]} \ldots \sum_{x_n \in Dom[V_n]}$$

**$P(\{V_1 = a, V_2 = x_2, V_3 = x_3, V_4 = x_4 \ldots, V_n = x_n\})$.**

# Probability Distributions over Finite Sets

Example:

$$P(\{V_1 = 1\}) = \sum_{x_2 \in Dom[V_2]} \sum_{x_3 \in Dom[V_3]} P(\{V_1 = 1, V_2 = x_2, V_3 = x_3\}).$$

| | | |
|---|---|---|
| (V1 = 1, V2 = 1, V3 = 1) | (V1 = 2, V2 = 1, V3 = 1) | (V1 = 3, V2 = 1, V3 = 1) |
| (V1 = 1, V2 = 1, V3 = 2) | (V1 = 2, V2 = 1, V3 = 2) | (V1 = 3, V2 = 1, V3 = 2) |
| (V1 = 1, V2 = 1, V3 = 3) | (V1 = 2, V2 = 1, V3 = 3) | (V1 = 3, V2 = 1, V3 = 3) |
| (V1 = 1, V2 = 2, V3 = 1) | (V1 = 2, V2 = 2, V3 = 1) | (V1 = 3, V2 = 2, V3 = 1) |
| (V1 = 1, V2 = 2, V3 = 2) | (V1 = 2, V2 = 2, V3 = 2) | (V1 = 3, V2 = 2, V3 = 2) |
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| (V1 = 1, V2 = 3, V3 = 1) | (V1 = 2, V2 = 3, V3 = 1) | (V1 = 3, V2 = 3, V3 = 1) |
| (V1 = 1, V2 = 3, V3 = 2) | (V1 = 2, V2 = 3, V3 = 2) | (V1 = 3, V2 = 3, V3 = 2) |
| (V1 = 1, V2 = 3, V3 = 3) | (V1 = 2, V2 = 3, V3 = 3) | (V1 = 3, V2 = 3, V3 = 3) |

# Probability Distributions over Finite Sets

Example:

$$P(\{V_1 = 1, V_3 = 2\}) = \sum_{x_2 \in Dom[V_2]} P(\{V_1 = 1, V_2 = x_2, V_3 = 2\}).$$

| | | |
|---|---|---|
| (V1 = 1, V2 = 1, V3 = 1) | (V1 = 2, V2 = 1, V3 = 1) | (V1 = 3, V2 = 1, V3 = 1) |
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| (V1 = 1, V2 = 3, V3 = 2) | (V1 = 2, V2 = 3, V3 = 2) | (V1 = 3, V2 = 3, V3 = 2) |
| (V1 = 1, V2 = 3, V3 = 3) | (V1 = 2, V2 = 3, V3 = 3) | (V1 = 3, V2 = 3, V3 = 3) |

# Joint Distributions

- A **joint distribution** over a set of random variables: $X_1, X_2, \ldots X_n$ specifies a real number for each assignment (or *outcome*):

$$P(X_1 = x_1, X_2 = x_2, \ldots X_n = x_n)$$

$$P(x_1, x_2, \ldots x_n)$$

  - Must obey:

$$P(x_1, x_2, \ldots x_n) \geq 0$$

$$\sum_{(x_1, x_2, \ldots x_n)} P(x_1, x_2, \ldots x_n) = 1$$

$P(T, W)$

| T | W | P |
|---|---|---|
| hot | sun | 0.4 |
| hot | rain | 0.1 |
| cold | sun | 0.2 |
| cold | rain | 0.3 |

**Note: Joint distribution can answer all probabilistic queries.**
**Problem: Table size is $d^n$.**

# Events

- An event is a set E of outcomes

$$P(E) = \sum_{(x_1 \ldots x_n) \in E} P(x_1 \ldots x_n)$$

- From a joint distribution, we can calculate the probability of any event
  - Probability that it's hot AND sunny? .4
  - Probability that it's hot? .4 + .1
  - Probability that it's hot OR sunny? .4 + .1 + .2

$$P(T, W)$$

| T | W | P |
|---|---|---|
| hot | sun | 0.4 |
| hot | rain | 0.1 |
| cold | sun | 0.2 |
| cold | rain | 0.3 |

General form

$$\Pr(X_i) = \sum_{X_1} \dots \sum_{X_{i-1}} \sum_{X_{i+1}} \dots \sum_{X_n} \Pr(X_1, X_2, \dots, X_n)$$

# Marginalization

- From a joint distribution (>1 variable) reduce it to a distribution over a smaller set of variables
- Called marginal distributions are sub-tables which eliminate variables
- Marginalization (summing out): Combine collapsed rows by adding likelihoods

$P(T, W)$

| T | W | P |
|---|---|---|
| hot | sun | 0.4 |
| hot | rain | 0.1 |
| cold | sun | 0.2 |
| cold | rain | 0.3 |

$$P(t) = \sum_s P(t, s)$$

$P(T)$

| T | P |
|---|---|
| hot | 0.5 |
| cold | 0.5 |

$$P(s) = \sum_t P(t, s)$$

$P(W)$

| W | P |
|---|---|
| sun | 0.6 |
| rain | 0.4 |

$$P(X_1 = x_1) = \sum_{x_2} P(X_1 = x_1, X_2 = x_2)$$

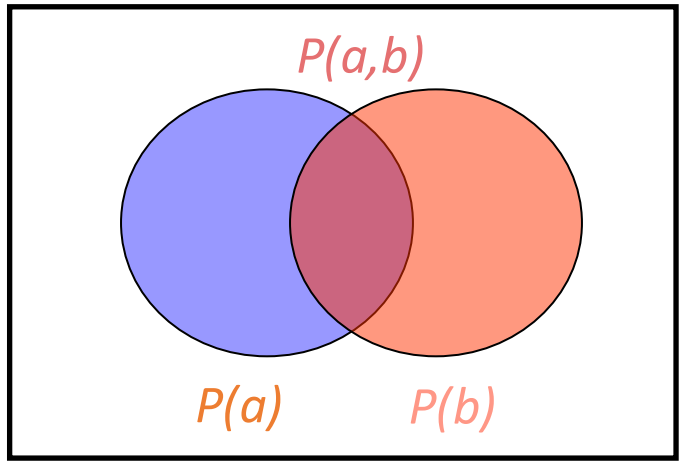
General form

$$Pr(X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n | X_i) = \frac{P(X_1,\dots,X_n)}{P(X_i)} = \frac{P(x_1,\dots,X_n)}{\sum_{X_1} \dots \sum_{X_{i-1}} \sum_{X_{i+1}} \dots \sum_{X_n} \Pr(X_1,\dots,X_n)}$$

# Conditioning

- Conditional distributions are probability distributions over some variables given fixed values of others

$$P(a|b) = \frac{P(a,b)}{P(b)}$$




$P(T, W)$ Joint Distribution

| T | W | P |
|---|---|---|
| hot | sun | 0.4 |
| hot | rain | 0.1 |
| cold | sun | 0.2 |
| cold | rain | 0.3 |

Conditional Distributions

$P(W|T)$

$P(W|T = cold)$

| W | P |
|---|---|
| sun | 0.4 |
| rain | 0.6 |

$P(W|T = hot)$

| W | P |
|---|---|
| sun | 0.8 |
| rain | 0.2 |

# Inference by Enumeration

- P(W)?

| S | T | W | P |
|---|---|---|---|
| summer | hot | sun | 0.30 |
| summer | hot | rain | 0.05 |
| summer | cold | sun | 0.10 |
| summer | cold | rain | 0.05 |
| winter | hot | sun | 0.10 |
| winter | hot | rain | 0.05 |
| winter | cold | sun | 0.15 |
| winter | cold | rain | 0.20 |

# Inference by Enumeration

- P(W)?

P(sun)=.3+.1+.1+.15=.65

| S | T | W | P |
|---|---|---|---|
| summer | hot | sun | 0.30 |
| **summer** | **hot** | **rain** | **0.05** |
| summer | cold | sun | 0.10 |
| **summer** | **cold** | **rain** | **0.05** |
| winter | hot | sun | 0.10 |
| **winter** | **hot** | **rain** | **0.05** |
| winter | cold | sun | 0.15 |
| **winter** | **cold** | **rain** | **0.20** |

# Inference by Enumeration

- P(W)?

P(sun)=.3+.1+.1+.15=.65
P(rain)=1-.65=.35

| S | T | W | P |
|---|---|---|---|
| summer | hot | sun | 0.30 |
| summer | hot | rain | 0.05 |
| summer | cold | sun | 0.10 |
| summer | cold | rain | 0.05 |
| winter | hot | sun | 0.10 |
| winter | hot | rain | 0.05 |
| winter | cold | sun | 0.15 |
| winter | cold | rain | 0.20 |

# Inference by Enumeration

- P(W | winter, hot)?

| S | T | W | P |
|---|---|---|---|
| summer | hot | sun | 0.30 |
| summer | hot | rain | 0.05 |
| summer | cold | sun | 0.10 |
| summer | cold | rain | 0.05 |
| winter | hot | sun | 0.10 |
| winter | hot | rain | 0.05 |
| winter | cold | sun | 0.15 |
| winter | cold | rain | 0.20 |

# Inference by Enumeration

- P(W | winter, hot)?

P(sun|winter,hot)~.1
P(rain|winter,hot)~.05
P(sun|winter,hot)=2/3
P(rain|winter,hot)=1/3

| S | T | W | P |
|---|---|---|---|
| summer | hot | sun | 0.30 |
| summer | hot | rain | 0.05 |
| summer | cold | sun | 0.10 |
| summer | cold | rain | 0.05 |
| winter | hot | sun | 0.10 |
| winter | hot | rain | 0.05 |
| winter | cold | sun | 0.15 |
| winter | cold | rain | 0.20 |

# Chain Rule

Chain rule is derived by successive application of product rule:

$$
\begin{aligned}
& P(X_1, \ldots, X_n) = \\
= \quad & P(X_1, \ldots, X_{n-1}) P(X_n | X_1, \ldots, X_{n-1}) \\
= \quad & P(X_1, \ldots, X_{n-2}) P(X_{n-1} | X_1, \ldots, X_{n-2}) P(X_n | X_1, \ldots, X_{n-1}) \\
= \quad & \ldots \\
= \quad & \prod_{i=1}^{n} P(X_i | X_1, \ldots, X_{i-1})
\end{aligned}
$$

**Constructing a larger distribution by simpler distribution.**

# Bayes Rule



- Two ways to factor a joint distribution over two variables:

$$P(x, y) = P(x|y)P(y) = P(y|x)P(x)$$

- Dividing, we get:

$$P(x|y) = \frac{P(y|x)}{P(y)}P(x)$$

- Example: Diagnostic probability from causal probability:

$$P(\text{cause}|\text{effect}) = \frac{P(\text{effect}|\text{cause})P(\text{cause})}{P(\text{effect})}$$

- Usefulness
  - Lets us build one conditional from its reverse.
  - Often one conditional is difficult to obtain but the other one is simple.

# Independence

- Two variables are *independent* if:

$$\forall x, y : P(x, y) = P(x)P(y)$$

  - This says that their joint distribution *factors* into a product two simpler distributions
  - Another form: $\forall x, y : P(x|y) = P(x)$
  - We write: $X \perp\!\!\!\perp Y$
  - Examples
    - N-independent flips of a fair coin.
    - P(*bus-arrives-on-time* | *wearing-blue-shirt)* = P(*bus-arrives-on-time*)

n smaller distributions

$P(X_1)$

| | |
|---|---|
| H | 0.5 |
| T | 0.5 |

$P(X_2)$

| | |
|---|---|
| H | 0.5 |
| T | 0.5 |

$\vdots$

$P(X_n)$

| | |
|---|---|
| H | 0.5 |
| T | 0.5 |

$P(X_1, X_2, \ldots X_n)$

$2^n$

# Conditional Independence

- X and Y are conditionally independent given Z.
- Given Z , X does not add information about Y and vice versa.

$$P(x, y \mid z) = P(x \mid z)P(y \mid z)$$

$$P(x \mid z) = P(x \mid z, y)$$

$$P(y \mid z) = P(y \mid z, x)$$

Example:

- Variable A: Student 1 gets an A on the exam.
- Variable B: Student 2 gets an A on the exam.
- Variable C (Condition): The official difficulty level of the exam (e.g., Easy vs. Hard)

# Samples from a Distribution

- Sampling from given distribution
  - Step 1: Get sample *u* from uniform distribution over [0, 1)
  - Step 2: Convert this sample *u* into an outcome for the given distribution by having each outcome associated with a sub-interval of [0,1) with sub-interval size equal to probability of the outcome
- Utility? We should be able to sample from a CPT defining the probabilistic model.
- Next, we look at approaches for sampling from a Bayes Net.

*Example:*

| C | P(C) |
|---|---|
| red | 0.6 |
| green | 0.1 |
| blue | 0.3 |

$$0 \leq u < 0.6, \rightarrow C = red$$
$$0.6 \leq u < 0.7, \rightarrow C = green$$
$$0.7 \leq u < 1, \rightarrow C = blue$$

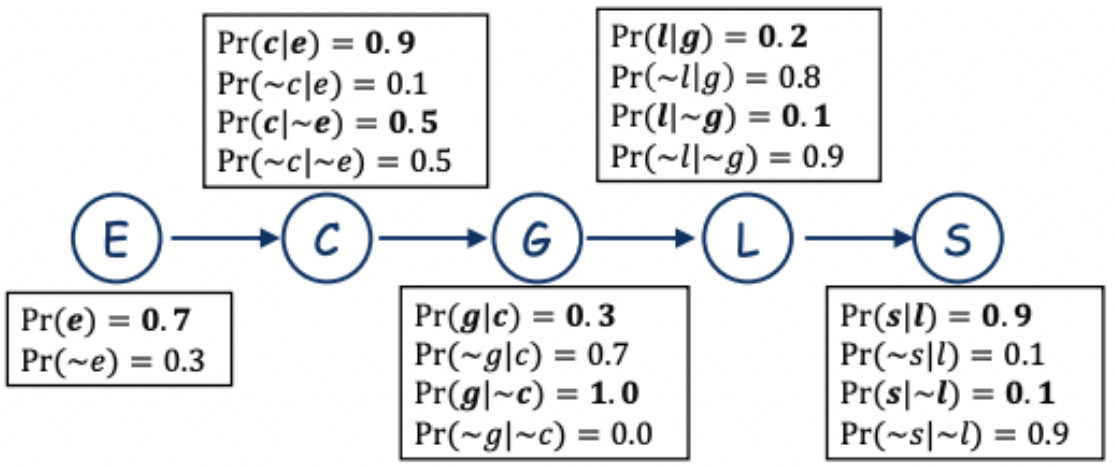
- If random() returns *u* = 0.83, then our sample is *C* = blue
- E.g, after sampling 8 times:

# Example: Sequence of probabilistic relationships

If Pascal woke up too early $E$, Pascal probably needs coffee $C$; if Pascal needs coffee, he's likely grumpy $G$. If he is grumpy then it's possible that the lecture won't go smoothly $L$. If the lecture does not go smoothly then the students will likely be sad $S$.




E - Pascal woke up too early  G - Pascal is grumpy  S - Students are sad
C - Pascal needs coffee  L- The lecture did not go smoothly



# Sometimes we get samples by simply taking actions

Follow a route and see how much time it takes to reach the goal.

Randomly play a game of chess.

Explore the frontier randomly while searching.

Sampling interactions (as we will see later) will lead to a certain form of "learning"

https://www.youtube.com/watch?v=TmPfTpjtdgg

model is a underlying mathematical rulebook of an environment

# Using samples to build estimates

Goal: Compute expected age of students enrolled in this class

we have the model

**Known P(A)** value iteration in the fully known MDP

$$E[A] = \sum_a P(a) \cdot a \quad = 0.35 \times 20 + \ldots$$

Without P(A), instead collect samples $[a_1, a_2, \ldots a_N]$

construct a estimated model

**Unknown P(A): "Model Based" way**

$$\hat{P}(a) = \frac{\text{num}(a)}{N}$$

$$E[A] \approx \sum_a \hat{P}(a) \cdot a$$

Eventually, we learn the right model which gives the right estimates.

bypass the model entirely

**Unknown P(A): "Model Free" way**

$$E[A] \approx \frac{1}{N} \sum_i a_i$$

Monte Carlo

Q learning

Bypass the model construction. Averaging works because the samples appear with the right frequencies.

# Takeaways

- Uncertainty is inherent in the real world.
- We want to (i) model uncertainty and (ii) incorporate the uncertainty in our decision making.
- This lecture is about modeling uncertainty – use of probabilities and random variables. Gives us tools to perform estimation.
- Important to think of the distribution over variables as a way to represent how entities we want to reason over are interacting with each other.
- Sometime we may not know about the distribution, we may only have samples from it. From samples we may need to estimate what underlying distribution the data emerged from.